PERSONAL NOTES AND FINDINGS: Exercises are at the bottom!
To start with Elliptic Curves it helps to have a geometric intuition of what they are to help strengthen our understanding.
An **An Elliptic Curve** is the set of solutions to:

$$\mathbb{Y}^2 = \mathbb{X}^3 + \mathbb{A}\mathbb{X} + \mathbb{B}$$

Two Examples include:

$$E : \mathbb{Y}^2 = \mathbb{X}^2 - 3\mathbb{X} + 3 \text{ and } \mathbb{Y}^2 = \mathbb{X}^2 - 6\mathbb{X} + 5$$

We are particularly interested in "adding" two points on elliptic curves to get a third point.
Definition 1. Let **P** and **Q** be two points on $\mathbb{E}$. We draw a line **L** through **P** and **Q**. This Line intersects $\mathbb{E}$ through three points. The third point we denote as **R**. We then take **R** and reflect it across the x-axis and get $\mathbf{R}'$, where $\mathbf{R}'$ is the sum of $\mathbf{P} + \mathbf{Q}$. This "addition" operation we denote with the symbol $\oplus$.

$$P \oplus Q$$

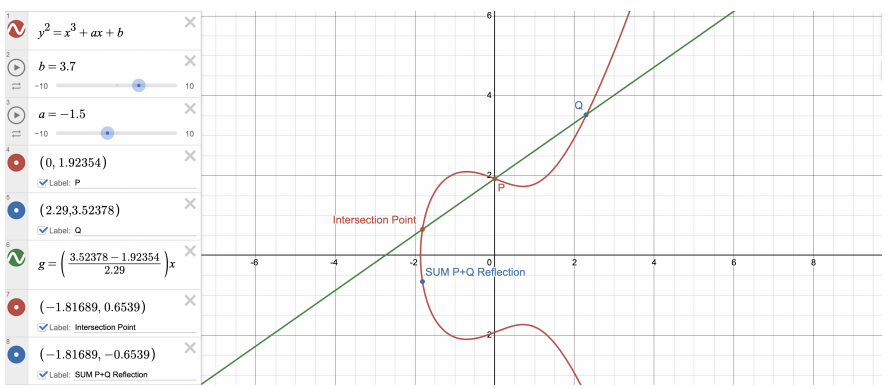

Figure 1: Notice a line Through P, Q and R' reflected across x-axis.

Example 0.1. Let E be an elliptic curve:

$$E : Y^2 = X^3 - 15X + 18$$

The points $P = (7, 16)$ and $Q = (1, 2)$ live on E. The line L connecting P and Q is given by

$$Y = \frac{7}{3}X - \frac{1}{3}$$

(We note that the slope $\frac{7}{3}$ is found with our standard $m = \frac{y_2 - y_1}{x_2 - x_1}$). In order to fine where E and L intersect we substitute and solve for X:

$$Y^2 = X^3 - 15X + 18 \implies \left(\frac{7}{3}X - \frac{1}{3}\right)^2 = X^3 - 15X + 18$$

$$\implies \left(\frac{49}{9}X^2 - \frac{14}{9}X + \frac{1}{9}\right) = X^3 - 15X + 18$$

$$\implies 0 = X^3 - \frac{49}{9}X^2 - \frac{121}{9}X + \frac{161}{9}$$

We already know two roots of this cubic polynomial (7 and 1) because P and Q are in the intersection of E and L.

$$X^3 - \frac{49}{9}X^2 - \frac{121}{9}X + \frac{161}{9} = (X-7)(X-1)\left(X + \frac{23}{9}\right)$$

$$\implies \left(X + \frac{23}{9}\right) \implies X = -\frac{23}{9}$$

We find the Y coordinate by:

$$Y = \frac{7}{3}X - \frac{1}{3} \implies Y = \frac{7}{3}\left(-\frac{23}{9}\right) - \frac{1}{3} \implies Y = \frac{170}{27}$$

Where now we have the sum of:

$$R = \left(-\frac{23}{9}, \frac{170}{27}\right) \implies R' = \left(-\frac{23}{9}, -\frac{170}{27}\right)$$

Therefore:

$$P \oplus Q = R' = \left(-\frac{23}{9}, -\frac{170}{27}\right)$$

We just handled the basic case of P and Q being separate points on the curve, but what happens if we add P to itself?? In this case we are to notice that L is a tangent line to the point P.

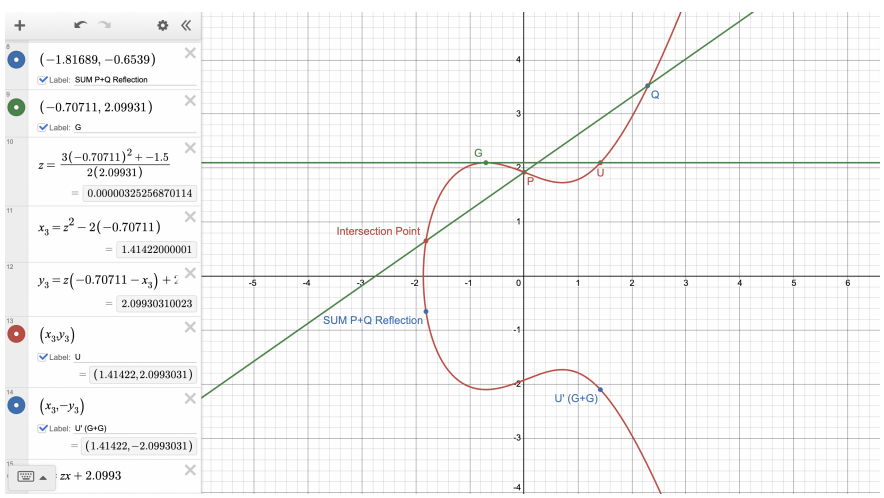

Figure 2: Notice a Tangent line Through G, and U reflected across x-axis to yield U' which is 2G.

$$2P = P \oplus P = R'$$

Example 0.2. Compute $P \oplus P$ from the previous elliptic curve:
The slope E at P is found by implicitly differentiating:

$$Y^2 = X^3 - 15X + 18$$

$$\implies 2Y\frac{dY}{dX} = 3X^2 - 15$$

$$\frac{dY}{dX} = \frac{3X^2 - 15}{2Y}$$

Subbing $P = (7, 16)$ gives the slope of $\lambda = \frac{33}{8}$. We can write the tangent line at P by:

$$L : Y = \frac{33}{8}X - \frac{103}{8}$$

We sub. Y back in to yield:

$$\left(\frac{33}{8}X - \frac{103}{8}\right)^2 = X^3 - 15X + 18$$

$$\implies X^3 - \frac{1089}{64}X^2 + \frac{2919}{32}X - \frac{9457}{64} = 0$$

7 is a double root of the cubic polynomial above:

$$(X - 7)\left(X - \frac{193}{69}\right) = 0$$

$$\implies \left(X - \frac{193}{69}\right) \implies X = \frac{193}{69}$$

$$Y = \frac{33}{8}\left(\frac{193}{69}\right) - \frac{103}{8} = -\frac{223}{512}$$

$$P \oplus P = \left(\frac{193}{64}, \frac{223}{512}\right)$$

We have our final case. What if we have point $P = (a, b)$ to it's x-axis reflection $P' = (a, -b)$ The line L would be a vertical line through P, P'. Call $L : x = a$.

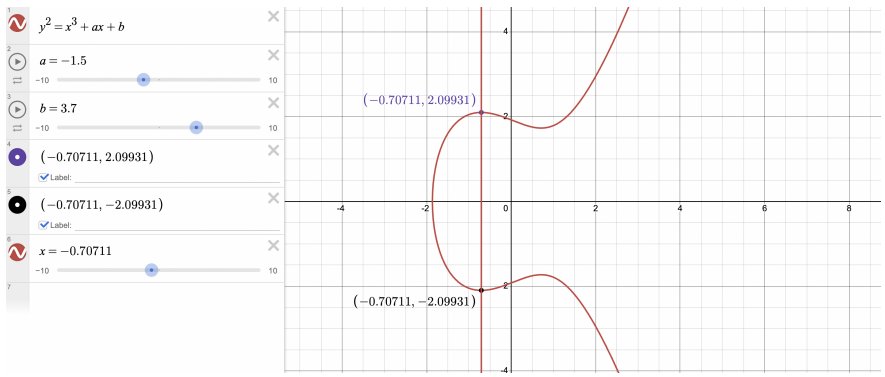

Figure 3: Notice a vertical line Through P, and its x-axis reflection which also intersects some point $\mathcal{O}$.

Our solution is to create some point $\mathcal{O}$.

$$P \oplus P' = \mathcal{O}$$

Additionally adding this point $\mathcal{O}$ to some ordinary point $P = (a, b)$ on E. L is the vertical line P and $\mathcal{O}$

$$P \oplus \mathcal{O} = P$$

Notice that $\mathcal{O}$ acts like an identity element, or zero on an elliptic curve.
Definition 2. An Elliptic Curve E is the set of solutions to a *Weierstrass* equation

$$E : Y^2 = X^3 + AX + B,$$

together, with an extra point $\mathcal{O}$. With A, B satisfying

$$4A^3 + 27B^2 \neq 0$$

Where the above term is the **discriminant** E. Which is equivalent to the condition that $X^3 + AX + B$ has no repeated roots.

We keep on talking about addition of points on these Elliptic Curves, so lets lay out a framework or "law" for this addition on E.

Theorem 3. *Let E be elliptic curves, then the addition law on E has the following:*

- *Identity:* $P + \mathcal{O} = \mathcal{O} + P = P$
- *Inverse:* $P + (-P) = \mathcal{O}$
- *Associative:* $(P + Q) + R = P + (Q + R)$
- *Commutative:* $P + Q = Q + P$

This addition law makes it clear that E is an ***abelian group***

This addition law combined with our concrete examples from before lays the groundwork for the ***Elliptic Curve Addition Algorithm***.

Theorem 4. *(The Elliptic Curve Addition Algorithm)*
Take the following Arbitrary Elliptic Curve:

$$E : Y^2 = X^3 + AX + B$$

Where P_1 and P_2 are points on E.

- *If $P_1 = \mathcal{O} \implies P_1 + P_2 = P_2$*
- *If $P_2 = \mathcal{O} \implies P_1 + P_2 = P_1$*
- *Otherwise we take $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$*
- *If $x_1 = x_2$ and $y_1 = -y_2$ then we know that P_2 is just the point P_1 reflected across the x-axis, thus $P_1 + P_2 = \mathcal{O}$*
- *Otherwise we define the slope of L by λ where:*

$$\lambda = \begin{cases} \frac{y_2 - y_1}{x_2 - x_1}, P_1 \neq P_2 \\ \\ \frac{3x_1^2 + A}{2y_1}, P_1 = P_2 \end{cases}$$

Let: $x_3 = \lambda^2 - x_1 - x_2$ and $y_3 = \lambda(x_1 - x_3) - y_1$ Thus

$$P_1 + P_2 = (x_3, y_3)$$

Omit the proof for now...

Now that we have built our geometric intuition on Elliptic curves we can start to transition into elliptic curves over the finite field $\mathbb{F}_{\text{ı}}$. It turns out that elliptic curves in this context are more useful for cryptography. This brings us to...

ELLIPTIC CURVES OVER FINITE FIELDS: Define an Elliptic curve over $\mathbb{F}_p$ by:

$$E : Y^2 = X^3 + AX + B, \text{ where } A, B \in \mathbb{F}_{\text{ı}} \text{ and } 4A^3 + 2B^2 \neq 0$$

And then we look at the points on E with coordinates in $\mathbb{F}_{\text{ı}}$

$$E(\mathbb{F}_p) = \{(x, y) : x, y \in \mathbb{F}_p \text{ and } y^2 = x^3 + ax + b\} \cap \{\mathcal{O}\}$$

Example 0.3. Consider the elliptic curve:

$$E : Y^2 + 3X + 8, \text{ over the field, } \mathbb{F}_{13}$$

We can find the points on $E(\mathbb{F}_{13})$ by subbing in all points for $X = (0, 1, 2, 3, ..., 12)$, checking to see if X is a square $\pmod{13}$. For example 8 is not a square

(mod 13). Lets try the point $X = 1$, which yields $1 + 3(1) + 8 = 12$, we know that $5^2 \equiv 12 \pmod{13}$ and $8^2 \equiv 12 \pmod{13}$. This gives us points $(1, 5)$ and $(1, 8)$ in $E(\mathbb{F}_{13})$. Continuing this process with all points we find that

$$E(\mathbb{F}_{13}) = \{\mathcal{O}, (1,5), (1,8), (2,3), (2,10), (9,6), (9,7), (12,2), (12,11)\}$$

Well what about addition of points in $E(\mathbb{F}_p)$? :
Take two points $P = (x_1, y_1), Q = (x_2, y_2) \in E(\mathbb{F}_p)$ We define $P + Q = (x_3, y_3)$ by applying the *Elliptic Curve Algorithm* . Since the coefficients of the elliptic curve belong to $\mathbb{F}_p$, then we end up with (x_3, y_3) whose coordinates also belong to $\mathbb{F}_p$.

Theorem 5. *Let E be an elliptic curve over $\mathbb{F}_p$, Let $P, Q \in E(\mathbb{F}_p)$*

- *The Addition Algorithm for P, Q yields a point in $E(\mathbb{F}_p)$*
- *The addition law on $E(\mathbb{F}_p)$ satisfies the addition law properties, thus making $E(\mathbb{F}_p)$ a finite group.*

Example 0.4. Take the curve: $E : Y^2 = X^3 + 3X + 8$ over $(\mathbb{F}_{13})$ Recall the points on this elliptic curve are:

$$E(\mathbb{F}_{13}) = \{\mathcal{O}, (1,5), (1,8), (2,3), (2,10), (9,6), (9,7), (12,2), (12,11)\}$$

Pick two points $P = (9, 7)$ and $Q = (1, 8)$ from $E(\mathbb{F}_{13})$. Calculate the slope of line L

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1}{-8} = \frac{1}{5} = 8$$

(We note that $\frac{1}{5} \pmod{13}$ amounts to $5x \equiv 1 \pmod{13}$ where the inverse of 5 is 8) Finally using the Addition Algorithm we find $x_3 = 64 - 9 - 1 = 54 \equiv 2 \pmod{13}$ and $y_3 = 8(9 - 2) - 7 = 49 \equiv 10 \pmod{13}$.

$$P + Q = (2, 10) \in E(\mathbb{F}_{13})$$

It is clear that the points in $E(\mathbb{F}_p)$ is a finite set as there are a finite number of possibles for X, Y coordinates. We see that $E(\mathbb{F}_p)$ is finite, *but how many elements live inside this set?*

The equation $Y^2 = X^3 + AX + B$ Over $E(\mathbb{F}_p)$ where there are p possibilities for X shows us that there are at most two values for Y for each value of X. Adding the extra point of $\mathcal{O}$ gives us $\#E(\mathbb{F}_p)$ is at most $(2p + 1)$.
When we plus X into $X^2 + AX + B$, there are 3 cases to consider:

- X may be a square $\pmod{p}$ in which is has 2 square roots, and we get two points $E(\mathbb{F}_p)$ which happens $\sim 50\%$ of the time.

- X may be a **non-residue** $\pmod p$ where we just discard that value for X. $\sim 50\%$ of the time.
- X might be zero, where we have 1 point in $E(\mathbb{F}_p)$. *THIS IS RARE.*

We now have that the approximate number of elements in $E(\mathbb{F}_p)$:

$$\sim \#E(\mathbb{F}_p) = 50\% \cdot 2p + 1 = p + 1$$

Which brings us to the following theorem:
Theorem 6. *(Hasse Theorem): Let E be a curve over $\mathbb{F}_p$.*

$$\#E(\mathbb{F}_p) = p + 1 - t_p \text{ where } |t_p| \leq 2\sqrt{p}$$

Elliptic Curve Discrete Log Problem: For the *DLP* over $\mathbb{F}_p^*$, Alice publishes g, h and her secret is x such that:

$$g^x \equiv h \pmod p$$

. To generalize the discrete log problem to groups, we are required to find some x such that

$$h \equiv g \cdot g \cdot g \cdot g ... (x - times) \pmod p$$

Now in the context of Elliptic curves Alice is working with the group of points $E(\mathbb{F}_p)$. Alice chooses and publishes $P, Q \in E(\mathbb{F}_p)$, and has some secret n that makes

$$Q = P + P + P + P ... (n - times) = nP \leftarrow n \text{ additions on E}$$

Definition 7. Let E be an elliptic curve over hte finite field $\mathbb{F}_p$ and let P, Q be points in $E(\mathbb{F}_p)$ The *Elliptic curve Discrete log problem* is finding $n \in \mathbb{Z}$ such that $Q = nP$

$$n = \log_p(Q)$$

It is important to note that similarly to the regular *DLP*, if we have some solution n then n is not unique, namely if n is a solution then $n = n_0 + is$ for $i \in \mathbb{Z}$ and s is the order of P: $Ps = \mathcal{O}$.

Proof. We know that $E(\mathbb{F}_p)$ is a finite group: If $P \in E(\mathbb{F}_p)$ then P is of finite order s, namely $Ps = \mathcal{O}$. Let n_0 be a solution to $Q = nP = n_0P$. Therefore:

$$nP = n_0P \implies nP - n_0P = \mathcal{O} \implies P(n - n_0) = \mathcal{O}$$

We know that $n - n_0$ but be a multiple of s thus

$$si = n - n_0 \implies n = n_0 + is$$

$\square$

We have found that $\log_p(Q)$ is an element of $(\mathbb{Z}/\sim\mathbb{Z})$, for s is the order of P. We can define the map $\log_p$ as a group homomorphism, because similarly to the standard DLP:

$$\log_p : E(\mathbb{F}_p) \to \mathbb{Z}/s\mathbb{Z}$$

$$\log_p(Q_1 + Q_2) = \log_p(Q_1) + \log_p(Q_2)$$

Example 0.5. Consider: $Y^2 = X^3+8X+7$ over $\mathbb{F}_{73}$. Take the points $P(32, 54)$ and $Q(39, 17)$ both in $E(\mathbb{F}_{73})$. We can verify using the Addition Algorithm on the point P that $Q = 11P$ and $\log_p(Q) = 11$.

Diffie-Hellman Key Exchange With Elliptic Curves: Alice and Bob agree to an Elliptic Curve over $\mathbb{F}_p$, $E(\mathbb{F}_p)$, and some point $P \in E(\mathbb{F}_p)$. Alice chooses some secret n_A and Bob chooses some secret n_B. They then compute the following:

$$(Q_A = n_A P) \text{ and } (Q_B = n_B P)$$

Alice and Bob exchange these values. Alice then uses her secret n_A to compute $n_A Q_B$ and Bob similarly computes $n_b Q_A$. ***They now have their shared secret value:***

$$n_A Q_B = n_A n_B P = n_B QA$$

Example 0.6. Alice and Bob choose the following variables $p = 3851$ and the curve $E : Y^2 = X^3 + 324X + 1287$, with the point $P(920, 303) \in E(\mathbb{F}_{3851})$. They then choose their respective secret values of: Alice - $n_A = 1194$ Bob - $n_B = 1759$ Alice computes the following:

$$Q_A = 1194P = (2067, 2178) \in E(\mathbb{F}_{3851})$$

Bob computes the following:

$$Q_B = 1759P = (3684, 3125) \in E(\mathbb{F}_{3184})$$

After exchanging these two values, Alice and Bob compute:

$$n_a Q_B = 1194(3684, 3125) = (3347, 1242) \in E(\mathbb{F}_{3851})$$

$$n_a Q_A = 1759(2067, 2178) = (3347, 1242) \in E(\mathbb{F}_{3851})$$

Alice and Bob's secret pair is $(3347, 1242)$ where they use the x value, 3347, as the secret key value.

Currently Alice and Bob are performing this process by changing points on $E(\mathbb{F}_p)$ where each point P contains two parts x_P and y_P. However if Alice and Bob wish to send each other fewer bits they realize that there is little necessity to send both coordinates over.
We know that x_P and y_P are related by:

$$Y_P^2 = X_p^3 + AX_P + B$$

There is not much additional information given by also sending the y-coordinate since that can be solved for by each respective sender.
If Alice decides to send Bob only the x-coordinate of Q_A Bob then can compute and choose 2 possible y-coordinates. In any-case Bob ends up computing

$$\pm n_B Q_A = \pm(n_A n_B)P$$

. Alice computes from her end similarly. Alice and Bob use the x-coordinate as their shared secret value, thus the y-coordinate is not fully essential.

Example 0.7. Alice and Bob use the following public parameters $p = 3851$, $E : Y^2 + 324X + 1287$ and point $P = (920, 303) \in E(\mathbb{F}_p)$. For their private values Alice and Bob choose $n_A = 2489$ and $n_B = 2286$. Where now Alice and Bob wish to send each other fewer bits. Alice and Bob compute the following

$$Q_A = n_A P = 2489(920, 303) = (593, 719)$$

$$Q_B = n_B P = 2286(920, 303) = (3681, 612)$$

Alice sends Bob $x_{Q_A} = 593$, Bob sends Alice $x_{Q_B} = 3681$. Alice uses 3681 to calculate:

$$y^2 = 3681^3 + (323)(3681) + 1287 \equiv 997 \pmod{3851}$$

$$\implies y^2 \equiv 997 \pmod{3851}$$

From our book we recall that Proposition 2.27 gave us $b^{(p+1)/4}$ is a square root of $b \pmod p$. Alice calculates:

$$y_B = 997^{3852)/4} = 997^{963} \equiv 612 \pmod{3851}$$

It happens that Alice gets the point $(3681, 612)$ as Q_B. She then computes $n_A Q_B = 2489(3681, 612) = (509, 1108)$.

In a similar fashion Bob runs through the following calculations:

$$y^2 = (593)^3 + 324(593) + 1287 \equiv 927 \pmod{3851}$$

$$\implies y_A = 927^{3852/4} = 927^{963} \equiv 3132 \pmod{3851}$$

$$Q'_A = (593, 3132)$$

$$n_B Q_A = 2286(593, 3132) = (509, 2743)$$

Bob and Alice share the secret key which is the x-coordinate of 509.

5.1 *Let $E : Y^2 = X^3 - 2X + 4$ be an elliptic curve with points $P = (0, 2)$ and $Q = (3, -5)$*

- $P + Q$ use:

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-7}{3}$$

$$x_3 = \left(\frac{-7}{3}\right)^2 - 0 - 3$$

$$y_3 = \lambda(x_1 - x_3) - y_1 \implies -\frac{7}{3}\left(0 - \frac{49}{9} - 3\right) - 2$$

$$P + Q = (x_3, -y_3) = \left(\left(\frac{-7}{3}\right)^2 - 0 - 3, -\left(-\frac{7}{3}\left(0 - \frac{49}{9} - 3\right)\right)\right)$$

$$P + Q = \left(\frac{22}{9}, -\frac{532}{27}\right)$$

- $P + P$:

$$\lambda = \frac{3x_1^2 + A}{2y} \implies \lambda = \frac{3(0) - 2}{2(2)} = \frac{-2}{4} = \frac{-1}{2}$$

$$x_3 = \frac{1}{4} - 0 - 2$$

$$y_3 = -\frac{1}{2}\left(0 - \frac{7}{4}\right) - 2$$

$$P + P = (x_3, y_3) = \left(\frac{1}{4} - 0 - 2, -\left(-\frac{1}{2}\left(0 - \frac{7}{4}\right) - 2\right)\right)$$

$$P + P = \left(-\frac{7}{4}, \frac{9}{8}\right)$$

- $Q + Q$:

$$\lambda = \frac{3x_1^2 + A}{2y} \implies \frac{3(3)^2 - 2}{2(-5)} = \frac{25}{-10}$$

$$x_3 = \left(\frac{25}{-10}\right)^2 - 3 - 3$$

$$y_3 = \frac{25}{-10}(3 - \left(\frac{25}{-10}\right)^2 - 3 - 3) + 5$$

$$Q{+}Q = (x_3, y_3) = \left(\left(\frac{25}{-10}\right)^2 - 3 - 3, -\left(\frac{25}{-10}(3 - \left(\frac{25}{-10}\right)^2 - 3 - 3) + 5\right)\right)$$

$$Q + Q = \left(\frac{1}{4}, -\frac{225}{8}\right)$$

5.3 Suppose that the Cubic Polynomial $X^3 + AX + B$ factors as

$$X^3 + AX + B = (X - e_1)(X - e_2)(X - e_3)$$

Prove that the discriminant $4A^3 + 27B^2 = 0$ i.f.f two or more e_1, e_2, e_3 are the same.

Proof. Take $X^3+AX+B = (X-e_1)(X-e_2)(X-e_3)$ and expand $(X-e_1)(X-e_2)(X-e_3)$ to yield:

$$(X - e_1)(X - e_2)(X - e_3)$$
$$\implies X^3 - e_3X^2 - e_2X^2 + e_2e_3X - e_1X^2 + e_1e_3X + e_1e_2X - e_1e_2e_3$$
$$\implies X^3 - (e_1 + e_2 + e_3)X^2 + (e_2e_3 + e_1e_2 + e_1e_3)X - e_1e_2e_3$$
$$X^3 + 0X^2 + AX + B = X^3 - (e_1 + e_2 + e_3)X^2 + (e_2e_3 + e_1e_2 + e_1e_3)X - e_1e_2e_3$$
$$0 = e_1 + e_2 + e_3$$
$$A = e_2e_3 + e_1e_2 + e_1e_3$$
$$B = -e_1e_2e_3$$

Therefore by algebraic manipulation we find that:

$$e_3 = -e_1 - e_2$$
$$A = e_2(-e_1 - e_2) + e_1e_2 + e_1(-e_1 - e_2)$$
$$-e_1^2 - e_1e_2 - e_2^2 = A$$

and

$$B = -e_1e_2(-e_1 - e_2) = e_1^2e_2 + e_1e_2^2$$
$$B = e_1e_2(e_1 + e_2)$$

We now arrive at

$$4A^3 + 27B^2 = -(e_1 - e_2)^3(e_1 - e_3)^2(e_2 - e_3)^2$$

This above result can be verified by plugging in the values of A and B and expanding. In order for $-(e_1 - e_2)^3(e_1 - e_3)^2(e_2 - e_3)^2 = 0$ a factor must be equal to zero. This is due to the fact that the product squares can only be zero if a factor is zero. Thus we find that

$$e_1 = e_2 \text{ or } e_1 = e_3 \text{ or } e_2 = e_3$$

Since we can verify above that

$$4A^3 + 27B^2 = -(e_1 - e_2)^3(e_1 - e_3)^2(e_2 - e_3)^2$$

, and know that the product of squares can only be zero if one of the squares is equal to zero itself we have finished. □

5.8 *Consider $E : y^2 = x^3 + x + 1$, let $P(4, 2)$ and $Q = (0, 1)$ be points on $E(\mathbb{F}_5)$ Solve the ECDLP for P, Q, find the integer n such that $Q = nP$*

Proof. Start with $P + P$: First we know that the slope

$$\lambda = \frac{3x_1^2 + A}{2y} \implies \frac{3(4)^2 + 1}{2(2)} \implies \frac{49}{9} = \lambda$$

Where since we are working with $\pmod 5$ we can reduce this to:

$$\frac{49}{9} \implies 49 \cdot 4^{-1} \pmod 5 \equiv 4 \cdot 4 \equiv 1 \pmod 5$$

Solving x_3 and y_3 now amounts to

$$x_3 = 1 - 4 - 4 = -7 \equiv 3 \pmod 5$$

$$y_3 = 1(4 + 7) - 2 = 9 \equiv 4 \pmod 5$$

$$2P = (3, 4)$$

Now solve $2P + P = (3, 4) + (4, 2)$:

$$\lambda = \frac{2 - 4}{4 - 3} = \frac{-2}{1} = -2 \cdot 1^{-1} \equiv 3 \pmod 5$$

$$x_3 = 3^2 - 3 - 4 = 2 \equiv 2 \pmod 5$$

$$y_3 = 3(3 - 2) - 4 \equiv -1 \equiv 4 \pmod 5$$

$$2P + P = (2, 4)$$

We can repeat this process until we eventually arrive at $P+P+P+P+P = 5P$ where $5P = (0, 1)$ thus $n = 5$. □

5.13 *Alice and Bob use the following $p = 2671$, $E : Y^2 = X^3 + 171X + 853$, and point $P(1980, 431) \in E(\mathbb{F}_{2671})$*

a.) Alice sends Bob $Q_A = (2110, 543)$. Bob decides on some value n_B so now he is able to compute the following:

$$Q_B = n_B P = (1943)(1980, 431) = (1432, 667)$$

Where Bob will just send Q_B back to Alice.

b.) Now in order to compute their shared secret value, we don't necessarily need to know Alice's secret value of n_A. Bob can just compute the following

$$n_B Q_A = 1943(2110, 543) = (2424, 911)$$

The secret key is the x-value of $n_B Q_A$, namely 2424.

c.) If the adversary Eve was to intercept the value of Q_A, in order to find the secret value of Alice's n_A she must iterate through additions of P until she can solve the ECDLP:

$$Q_A = n_A P = \implies (2110, 543) = (n_A)(1980, 431)$$

d.) If Alice sends Bob $x_A = 2$ from Q_A, and Bob decides on some $n_B = 875$. He can compute the following:

$$Y^2 = X^3 + 171X + 853 \implies (2)^3 + 171(2) + 853 = 1203$$

$$Y^2 = 1203 \implies Y = 1203^{2672/4} = 1203^{668} \equiv 2575$$

Now we have found $Q_A = (2, 2575)$ Where Bob computes his value of:

$$n_B Q_A = 875(2, 2575) = (1708, 1419)$$

. Thus their shared key is 1708 as we only pay attention to the x-value.